

The Effect of Zinc and Calcium In Yogurt, and How Long The Probiotic Survives

Presentation : Aerial M. Research

How zinc Gives a Positive Effects
On The probiotic Endurance In
The Digestive System

Materials

- 3 yogurts
 - Kefir
 - Yakult
 - L.Reuteri
 - Tube
 - Incubator
 - Agar plates
 - Polypropylene plastic (for coupons)
 - Well plate for serial dilution
 - Goggles-gloves-lab coat

Conclusion

- Our main goal was to see how much of the bacteria would grow and see how long they stayed alive in a beneficial dietary supplement.
- Yogurt (after 4 weeks in fridge):
 - Kefir cultured in milk alone has the most viable the bacteria while zinc didn't help kefir.
 - Yakult prefer calcium and zinc together (both promote more viable bacteria).
 - *L. reuteri* cultured in milk and calcium yield the most viable bacteria
- Kefir in milk (After 3 days):
 - Zn and Ca didn't help with the viability of the planktonic bacteria but Zn and Ca and both support stronger viability for biofilm.
- *L. reuteri* in CDM (day 1-3):
 - *L. reuteri* biofilm viability is improved with increased zinc concentration (25-100 μM) for day 1 and 25 μM Zn is the condition that yield most viable biofilm cells after 3 days.